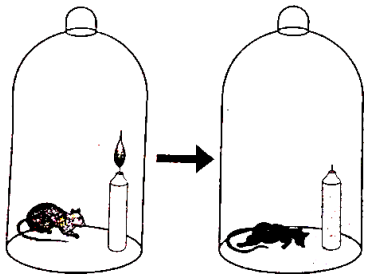
Biology

Topic: Photosynthesis in higher plants

What Do We know? and Early Experiments

1. Which of the following experiment showed that CO_2 is essential for photosynthesis?
1) Moll half-leaf experiment
2) Priestley's experiment
3) Inverted funnel experiment
4) Variegated leaf experiment

2. The following set-up is called



1) Engelmann experiment
2) Jan Ingenhousz experiment
3) Priestley's experiment
4) Calvin experiment

3. The first action spectrum of photosynthesis was described by__A_using __B__ and C .

	A	B	C
1)	T. W. Engelmann	Cladophora	Anaerobic bacteria
2)	T. W. Engelmann	Cladophora	Aerobic bacteria
3)	Jan Ingenhousz	Spirogyra	Aerobic bacteria
4)	Julius von Sachs	Chlorella	Anaerobic bacteria

4. _____ showed that in the presence of sunlight, only the green parts of the plants could release oxygen.
1) Ingenhousz
2) Van Niel
3) Priestley
4) Julius von Sachs

5. ____ found that glucose is produced in the green part of the plant.
1) Joseph Priestley
2) T. W. Engelmann
3) Julius Von Sachs
4) Jan Ingenhousz

6. Select the incorrect match.

1) T. W. Engelmann — Action spectrum
2) Blackman — Law of limiting factor
3) Van Neil — Enhancement effect
4) Melvin Calvin — C_3 pathway

7. The process of photosynthesis was first discovered by
1) Priestley
2) Ingenhousz
3) Engelmann
4) Blackman

8. What is the oxidation product when H_2S is used as the hydrogen source for photosynthesis in purple and green sulphur bacteria?
1) Oxygen
2) Sulphur
3) CO_2
4) Nitrogen

9. Match the given columns.

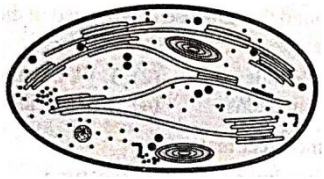
Column I (Scientist)		Column II (Experimental material)	
a	Joseph Priestly	1	Purple and green bacteria
b	T.W. Engelmann	2	Algae
c	Cornelius van Niel	3	Cladophora
d	Melvin Calvin	4	Mint plant

	a	b	c	d
1)	3	4	2	1
2)	1	2	3	4
3)	4	3	2	1
4)	4	3	1	2

Where Does Photosynthesis Take Place? How Many Pigments are Involved in Photosynthesis?

10. Read the following four statements (A-D) on photosynthesis.
A. Photosynthesis is an oxidoreductive process.
B. The stroma of chloroplast is responsible for the synthesis of ATP and NADPH.
C. Membrane system of chloroplast is responsible for photochemical reaction.
D. Photosynthesis does not take place in the ground tissue of leaves.
Select the right option having both incorrect statements.
1) (A) and (D)
2) (B) and (D)
3) (B) and (C)
4) (C) and (D)

11. Study the following figure and select the incorrect statement.



1) It shows clear-cut division of labour.
2) It has membranous system for trapping sunlight.
3) They align themselves along the wall of mesophyll cells.
4) It has light-independent non-enzymatic reactions for CO_2 fixation.

12. Select the correct equation for photosynthesis.

1) $6CO_2 + 12H_2O \xrightarrow{\text{Light}} C_6H_{12}O_6 + 6H_2O + 6O_2$
2) $6CO_2 + 6H_2O \xrightarrow{\text{Light}} C_6H_{12}O_6 + 12H_2O + 6O_2$
3) $12CO_2 + 6H_2O \xrightarrow{\text{Light}} C_6H_{12}O_6 + 6CO_2 + 12H_2O$
4) $6O_2 + 12H_2O \xrightarrow{\text{Light}} C_6H_{12}O_6 + 12H_2O + 12O_2$

13. Photosynthesis is an

1) Oxidoreductive process
2) Anabolic process
3) Endergonic process
4) All of these

14. Which metal ion is a constituent of chlorophyll?

1) Iron
2) Copper
3) Magnesium
4) Zinc

15. Match the following columns and choose the correct option.

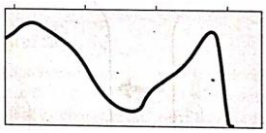
Column I		Column II	
(A)	Chlorophyll a	(i)	Yellow orange
(B)	Chlorophyll b	(ii)	Yellow
(C)	Xanthophyll	(iii)	Blue green
(D)	Carotene	(iv)	Yellow green

1) A-(iii), B(iv), C- (i), D- (ii)
2) A- (iii), B- (iv), C- (ii), D- (i)
3) A-(ii), B-(iii), C- (i), D- (iv)
4) A-(iv), B- (iii), C – (i), D – (ii)

16. The primary pigment of photosynthesis

1) Forms the reaction centre of photosystem
2) Forms the antenna molecule
3) Is chlorophyll a molecule
4) Both (1) and (3)

17. The following graph shows



1) Absorption spectrum of chlorophyll a
2) Absorption spectrum of chlorophyll b
3) Absorption spectrum of carotenoids
4) Action spectrum of photosynthesis

18. Which pigment acts directly to convert light energy to chemical energy?
1) Chlorophyll a
2) Chlorophyll b
3) Xanthophyll
4) Carotenoid

19. How can we separate the leaf pigments of any green plant?
1) Column chromatography
2) Paper chromatography
3) Spectrophotometer
4) Autoradiography

What is Light Reaction? and The Electron Transport

20. During light reaction in photosynthesis, the following are formed
1) ATP and sugar
2) Hydrogen, O_2 and sugar
3) ATP, hydrogen donor, and O_2
4) ATP, hydrogen, and O_2 donor

21. The very first event in photosynthesis is
1) Photoexcitation of chlorophyll molecule
2) Splitting of water molecule
3) Photophosphorylation
4) Reduction of $NADP^+$

22. Read the following statements and choose the incorrect option.
1) Several protein complexes are involved in the photochemical process.
2) PS I and PS II are named in the sequence in which they function during the light reaction, and not in the sequence of their discovery.
3) The reaction centre is different in both the photosystems i.e., PS I and PS II.
4) Each photosystem has all the pigments (except one molecule of chlorophyll a) forming light harvesting system.

23. Read the following four statements A, B, C, and D and select the correct option having both incorrect statements.
(A) Antennae molecules are made up of hundreds of pigment molecules bound to proteins.
(B) Antennae pigments help to make photosynthesis more efficient by absorbing different wavelengths of light.
(C) Only PS II is functional in cyclic photophosphorylation.
(D) In PS I, the reaction centre is chlorophyll a, which absorbs light of 680 nm.

1) (A) and (C)
2) (B) and (C)
3) (C) and (D)
4) (B) and (D)

24. Which of the following are formed during light reaction in C_3 plants?
1) Hydrogen, O_2, and sugar
2) ATP, reducing power, and oxygen
3) ATP, reducing power, and CO_2
4) Hydrogen, reducing power, and sugar

25. Chemiosmotic hypothesis
1) Was experimented by P. Mitchell.
2) Proves that proton gradient is required for ATP synthesis.
3) Explains how ATP is synthesised in chloroplast and mitochondria.
4) All of these.

26. Oxygen evolving complex
1) Is associated with the reaction centre of PS II
2) Is located on the inner side of thylakoid membrane
3) Requires Mn, Ca, and Cl
4) All of these

27. Light reactions in green plants include all, except
1) Photoexcitation of chlorophyll molecule
2) Water splitting reaction
3) Reduction of CO_2
4) Generation of ATP and $NADPH_2$

28. Z scheme is
(i) Cyclic photophosphorylation.
(ii) Non-cyclic photophosphorylation.
(iii) Proposed by P. Mitchell.
1) Only (i)
2) Only (ii)
3) Both (i) and (iii)
4) Both (ii) and (iii)

29. When the two photosystems work in a series, first PS II and then the PS I, which of the following process occurs?
1) Non-cyclic photophosphorylation
2) Cyclic photophosphorylation
3) Chemiosmosis
4) Dark reaction

30. The reaction centre of PS I is __A__ and the reaction centre of PS II is ___
1) P_{680}, P_{700}
2) P_{700}, P_{680}
3) P_{700}, P_{780}

4) P_{780}, P_{700}

Where are ATP and NADPH Used?

31. Dark reaction in photosynthesis is called so because
1) It can occur in dark also
2) It does not directly depend on light energy
3) It cannot occur during day light
4) It occurs more rapidly at night

32. The biosynthetic phase of photosynthesis
(A) Involves use of ATP and NADPH to produce sugar.
(B) Takes place in the stroma of chloroplast.
(C) Does not depend directly on the presence of light.
(D) Depends on the products of light reaction.
The correct statements are
1) Only (B) and (C)
2) Only (A), (B), and (C)
3) Only (A) and (C)
4) (A), (B), (C), and (D)

33. Where does the dark reaction of photosynthesis takes place?
1) Stroma of chloroplast
2) Thylakoid lumen
3) Thylakoid membrane
4) Thylakoid membrane

34. Reat the given statements.
Statement I: In fixation of six molecules of CO_2 in photosynthesis, 18 ATP and 12 NADPH are used to form one molecule of hexose sugar.
Statement II: The light reaction results in formation of ATP and NADPH.
1) Both statement I and statement II are true.
2) Both statement I and statement II are false.
3) Statement I is true and statement II is false.
4) Statement I is false and statement II is true.

35. C_3 plants require ____A____ ATP molecules and ____B____NADPH molecules for fixation of one CO_2 molecule.

	A	**B**
1)	18	12
2)	3	2
3)	2	3
4)	5	2

36. Melvin Calvin traced the path of CO_2 fixation using all except
1) Radioactive C^{14} in CO_2
2) Chlorella

3) Chromatography and autoradiography
4) Cladophora and aerobic bacteria

37. The enzyme that is not found in a C_3 plant is
1) RuBP Carboxylase
2) PEP Carboxylase
3) NADP reductase
4) ATP synthase

38. Match the column I with column II and choose the correct option (for C_3 plant).

Column I		Column II	
a	Primary CO_2 acceptor	1	RuBP
b	Primary CO_2 fixation product	2	PGA
c	Cell type in which the Calvin cycle takes place	3	Mesophyll
d	Cell type in which the initial carboxylation reaction occurs	4	Bundle sheath

1) a-1, b-2, c-3, d-4
2) a-1, b-2, c-4, d-3
3) a-2, b-1, c-3, d-3
4) a-1, b-2, c-3, d-3

39. Which of the following ratios is correct for producing one molecule of glucose through six rounds of Calvin cycle?

	CO_2	ATP	NADPH
1)	6	18	12
2)	6	12	18
3)	1	3	2
4)	1	2	3

40. Fill in the blanks in the given sentence.
Reduction phase of Calvin cycle involves a series of reactions that lead to the formation of glucose. The steps involve utilisation of ____ molecules of ATP for phosphorylation and ____ of NADPH for reduction per CO_2 molecule fixed.
1) 2,2
2) 4,2
3) 3,3
4) 2,4

C_4 Pathway and Photorespiration

41. Read the following statements on photorespiration and state them as true (T) or false (F).
(A) Oxidation of RuBP in the stroma of chloroplast in bright light and high oxygen.
(B) Does not utilise ATP and reducing power.

(C) Results in the release of CO_2 with the utilisation of energy.

(D) The biological function of photorespiration is not yet known.

	A	B	C	D
1)	F	T	T	F
2)	F	F	T	T
3)	T	F	T	F
4)	T	F	T	T

42. The number of ATP molecules consumed for the synthesis of one glucose molecule by rice and maize plants are, respectively,

1) 18,30
2) 18, 12
3) 18, 18
4) 30, 18

43. PEP is primary CO_2 acceptor in

1) C_4 plants
2) C_3 plants
3) C_2 plants
4) Both C_3 and C_4 plants

44. Match the column I with column II, and choose the correct option.

Column I (Compound)		**Column II (Number of carbons)**	
a	RuBP	1	Two
b	PEP	2	Three
c	PGA	3	Four
d	OAA	4	Five

1) a-3, b-4, c-2, d-1
2) a-1, b-2, c-4, d-3
3) a-4, b-2, c-2, d-3
4) a-3, b-2, c-2, d-4

45. High photosynthetic efficiency is found in

1) C_3 plants
2) C_4 plants
3) CAM plants
4) C_4 and CAM plants

46. How many plants given below have C_3 as well as C_4 pathways and mechanism that increase the concentration of CO_2 at the enzyme site during photosynthesis?

Wheat, maize, rice, potato, sugarcane, tomato, pearl millet, bell pepper, Sorghum

1) Three
2) Four
3) Six
4) Five

47. Consider the following statements on C_4 plants and select the correct option.

(A) They are adapted to dry temperate regions.

(B) They have a special type of leaf anatomy.
(C) They show response to high light intensities.
(D) They lack a process called photorespiration.
1) (A) and (C) are correct.
2) (A), (B), and (D) are correct.
3) (B), (C), and (D) are correct.
4) (A), (B), (C), and (D) are correct.

48. Read the following statements and choose the correct option.
Statement I: Plants possessing C_4-pathway of carbon fixation have a higher net primary productivity than the C_3-pathway possessing plants.
Statement II: For each unit weight of fixed carbon, C_4-pathway possessing plants require less water than the C_3- pathway possessing plants.
1) Both statement I and statement II are true.
2) Both statement I and statement II are false.
3) Statement I is true and statement II is false.
4) Statement I is false and statement II is true.

49. Photorespiration is favoured by
1) High O_2 and low CO_2
2) Low light and low O_2
3) Low temperature and low O_2
4) Low O_2 and high CO_2

50. Kranz anatomy is one of the features of the leaves of
1) C_3 plants only
2) C_4 plants only
3) C_4 and CAM plants
4) C_3 and C_4 plants

51. The correct sequence of cell organelles involved in photorespiration is
1) Chloroplast, peroxisomes, mitochondria
2) Peroxisome, mitochondria, chloroplast
3) Mitochondria, chloroplast, peroxisome
4) Chloroplast, mitochondria, peroxisome

52. Which of the following processes in mesophyll cell results in the release of CO_2 with the utilisation of ATP?
1) C_3 pathway
2) Photorespiration
3) C_2 pathway
4) Both (2) and (3)

Factors Affecting Photosynthesis

53. Major limiting factor for photosynthesis is
1) H_2O

2) CO_2
3) Light
4) Temperature

54. Which of these factors has no direct effect on photosynthesis?
1) Light
2) Water
3) Temperature
4) CO_2

55. Relationship between incident light and CO_2 fixation rate is
1) Sigmoid at low light intensity
2) Linear at low light intensity
3) Parabola at low light intensity
4) Both (2) and (3)

56. CO_2 compensation point for C_3 plant is
1) 25-100 ppm
2) 0-10 ppm
3) 10-20 ppm
4) 200-300 ppm

57. Light saturation occurs at
1) 10 per cent of the full sunlight
2) 1 per cent of the full sunlight
3) 0.1 per cent of the full sunlight
4) 50 per cent of the full sunlight

58. Read the following statements and state them as true (T) or false (F).
A. If CO_2 concentration increases up to 0.05%, the rate of photosynthesis increases for short term.
B. The C_3 and C_4 plants respond similarly to CO_2 concentration.
C. C_3 plants respond to high temperature and show higher photosynthetic rate.
D. Light reaction is less temperature sensitive than dark reaction.

	A	B	C	D
1)	T	F	F	T
2)	T	F	F	F
3)	F	T	T	F
4)	F	F	T	T

59. Current availability of CO_2 levels in atmosphere is limiting to all, except
1) Wheat and tomato
2) Tomato and rice
3) Maize and sugarcane
4) Tomato and bell pepper

60. Who gave the "law of limiting factors?
1) Emerson
2) Jan Ingenhousz
3) Blackman

4) Engelmann

61. CO_2 saturation point for C_3 plants (A) and C_4 plants (B) is

	A	B
1)	360 μL / L	450 μL / L
2)	450 μL / L	360 μL / L
3)	250 μL / L	150 μL / L
4)	360 μL / L	650 μL / L

62. Increase in light intensity beyond saturation point causes

1) More photosynthesis
2) No change in rate of photosynthesis
3) Breakdown of chlorophyll and a decrease in photosynthesis
4) Heliotropism

63. Read the following statements and choose the incorrect option.

1) At high light intensities, both C_3 and C_4 plants show increase in the rates of photosynthesis.
2) The C_3 plants respond to higher temperatures and show higher rate of photosynthesis while C_4 plants have a much lower temperature optimum.
3) Even though water is one of the reactants in the light reaction, the effect of water as a factor is more through its effect on the plant, rather than directly on photosynthesis.
4) The plant factors affecting photosynthesis include the number, size, age and orientation of leaves, mesophyll cells and chloroplasts, internal CO_2 concentration and the amount of chlorophyll.

64. C_3 plants respond to higher CO_2 concentration by showing increased rates of photosynthesis, leading to higher productivity. This feature has been used for some greenhouse crops such as

1) Maize and sugarcane
2) Wheat and sorghum
3) Tomato and sugarcane
4) Tomato and bell pepper

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
1	3	2	1	3	3	2	2	4	2	4	1	4	3	2
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
4	4	1	2	3	1	2	3	2	4	4	3	2	1	2
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
2	4	1	3	2	4	2	4	1	1	4	1	1	3	2
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
2	3	1	1	2	1	2	2	2	2	1	1	1	3	3
61	**62**	**63**	**64**											
2	3	2	4											